Peripheral Blood Smear Analysis Using Automated Computer-Aided Diagnosis System to Identify Acute Myeloid Leukemia

Vasundhara Acharya, Vinayakumar Ravi, Tuan D. Pham, *Senior Member, IEEE*, and Chinmay Chakraborty

***Abstract*—Acute myeloid leukemia (AML) is a cancer of the myeloid line of cells caused due to the rapid increase of abnormal cells that later interfere with healthy cells. One of the main reasons for the increase in mortality is the cost of the devices used for the determination and late diagnosis. The most effective treatment option can be provided by accurate medical diagnosis. Automated segmentation of blood smear images plays a crucial role in the identification of the AML. This article proposes a new computer-aided diagnosis model to segment the blood smear images and identifies the stage of AML. The methodology presented in this work consists of various stages: Image acquisition, image segmentation, feature extraction/selection, and classification. The model is trained using 800 blood smear images collected from Kasturba Medical College Manipal, and 200 images collected from the dataset of microscopic peripheral blood cell images for the development of automatic recognition systems. The model is tested on 500 images. A novel algorithm is designed to accurately segment the blood smear images to identify AML and its stages. The segmentation algorithm addresses critical issues in blast cell detection, including identifying the blast cell and extracting the cytoplasm of the cell without involving manual intervention. It can identify the multiple lobes and the nucleated red blood cells (NRBCs), separate the overlapped erythrocytes from white blood cells (WBCs), and discover the presence of Auer rods and granules. The feature selection is performed using the InfoGainAttributeEval and the ranker search method. This article compares the performance of the various machine learning algorithms exploited for the classification of different types of cells and hence determines AML. The model successfully differentiated between NRBC and WBC with an accuracy of 99.81%. The model obtained a classification accuracy of 99.48 %. It achieved a prediction accuracy of 99.2% while predicting the unknown stage of AML. The algorithm's efficiency proves that it can be used by pathologists to form a prognosis.**

***Index Terms*—Acute myeloid leukemia (AML), blood smear images, computer-aided diagnosis (CAD), nucleated red blood cells, white blood cells (WBC).**

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I. Introduction

Acute myeloid leukemia (AML) is a cancer of the blood or bone marrow. Examination of the peripheral blood smear images helps in identifying the disease. Various manual and automated techniques exist to analyze the blood smear images. The main downside of manual intervention is less precision and time consumption [1].

It is also prone to sampling error. Laser sensors are employed in the automated technique. Even though they obtain an accurate count of the number of cells, they fail in identifying the abnormal cells. Automation of the entire process reduces the burden on the pathologists and generates precise results. The development of the automated diagnostic tool to identify AML involves the following stages: Image acquisition, image segmentation, feature extraction/selection, and classification.

According to a survey, AML is the second most diagnosed leukemia in both children and adults [2]. The AML is classified into eight subtypes, namely, M0–M7 by the French–American–British (FAB) classification system. According to the hematologists, M2–M5 is found to be highly prevalent [3].

Granulopoiesis is a procedure by which granulocytes are produced. It is a type of white blood cell with multilobed nuclei. Myeloblast, myelocyte, and promyelocyte are generated in the process of granulopoiesis [4]. Auer rods are red-stained needles like structures that are found in the cytoplasm of myeloblasts or progranulocytes. Its presence helps in diagnosing nonlymphocytic (myeloid) leukemia. Monocytopoiesis [5] is a procedure by which monocytes are generated. Monoblast and promonocyte are produced in this process. AML-M2 type is characterized by the presence of myeloblasts with maturation. The presence of promyelocyte denotes AML-M3 type. AML-M4 type is signified by the presence of myeloblasts, myelocytes, promonocytes, and Monoblasts. The presence of Monoblasts indicates AML-M5 type. Many hematology analyzers misclassify NRBC and produce a wrong total count of WBCs as they have a size and nucleus similar to that of WBCs [6]. In the proposed work, these cells are identified and differentiated from the WBC, thereby correcting the count. The percentage count of each type of these cells is used in categorizing the disease into its stages. The different types of cells are depicted in Fig. 1.

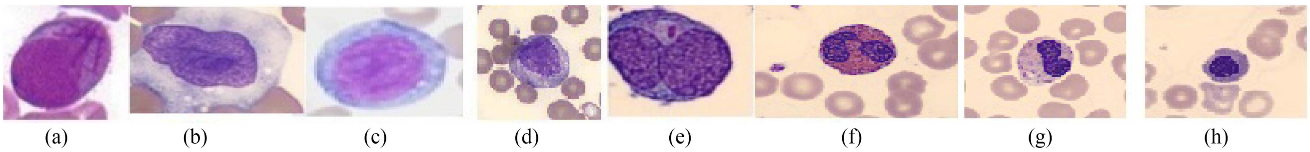


Fig. 1. (a) Myeloblast. (b) Promonocyte. (c) Monoblast. (d) Myelocyte. (e) Promyelocyte. (f) Eosinophil. (g) Neutrophil. (h) NRBC.

II. Related Works

Various research works have been carried out in the field of blood smear segmentation to its components and the detection of acute leukemia types. Machine learning and deep learning have proven to trim the time it takes to review patient and medical data, leading to faster diagnosis and speedier patient recovery [7], [8]. Abdeldaim *et al.* [3] designed a system to diagnose acute lymphoblastic leukemia (ALL) by segmenting the WBCs using the color conversion and Zack's thresholding technique. Alomari *et al.* [9] used an iterative circle detection algorithm to count WBCs and RBCs. A threshold value of 64 was used to segment WBCs from the RBCs. Khashman *et al.* [10] designed a model to identify ALL by segmenting the WBCs using Otsu's threshold. An averaging using a 2*2 kernel was used to "fuzzify" the images. Putzu and Ruberto [11] proposed an approach to segment the WBCs from other components in the blood using the threshold value computed using Zack's algorithm. The resulting image was cropped to obtain one WBC per image. The cytoplasm and nucleus of WBC were later extracted from the cropped image. Ruberto *et al.* [12] proposed a technique to segment the WBCs using the support vector machine (SVM). The WBCs were counted by employing the circular Hough transform. Sarrafzadeh *et al.* [13] proposed a model to segment WBCs from the smear using the K-means algorithm and the initial seed points. Jiang *et al.* [14] used scale-space filtering to extract the nucleus region from the subimage. The cytoplasm was extracted using watershed clustering in 3D HSV (hue, saturation, value) histogram. Guo *et al.* [15] applied the SVM) to the spectrum of each pixel present in the bone microscopic images to achieve segmentation. Ahasan [16] developed an algorithm to diagnose leukemia by extracting WBCs from the peripheral blood film using color space conversion and color filtering. Madhloom *et al.* [17] proposed a method to separate the lymphoblasts present in the smear from the other cells using the combination of color features with morphological reconstruction technique. Mishra *et al.* [18] developed a model to diagnose ALL by obtaining the WBCs using marker-based segmentation. Hegde *et al.* [19] [20], developed a robust algorithm to diagnose the WBC nuclei and leukocytes. The normalized and contrast-enhanced G component of the image was used. The TissueQuant method was implemented to select the color and shade efficiently. Osowski [21] proposed a technique to segment bone marrow images using the watershed transform, and feature selection were performed using the genetic algorithm (GA). Ghane *et al.* [22] proposed a model to segment the WBCs and their components using the K-means clustering algorithm. The mask of the WBC present in the smear was obtained after applying thresholding and morphological operations. Sudha *et al.* [23] proposed a technique to segment the leukocytes using an *n* edge strength-based Grabcut method. The WBCs were then counted using the gradient circular hough transform method. Wang *et al.* [24] designed a model to diagnose the nucleus of the leukocyte by enhancing the saliency of the saturation component. The resulting image was converted to binary using Otsu's threshold. The empirical value of the small area was utilized to remove platelets and other artifacts to retain only the nucleus. Hegde *et al.* [25] proposed a technique to extract the nuclei by image enhancement method. The extracted features were fed into the neural network for classification. Fan *et al.* [26] presented a technique to localize and segment WBC using deep neural networks. Pixel-level information was used to train the model. It was later used to locate the leukocyte and segment the mask of the leukocyte. Wang *et al.* [27] considered leukocyte recognition as an object detection task and applied Single Shot Multibox Detector and You Only Look Once to detect the objects. Hegde *et al.* [28] presented a summary of the methods used to analyze the blood smear image. The work categorized the methods into three groups based on WBC analysis, RBC analysis, and platelet analysis. Neoh *et al.* [29] proposed a clustering algorithm with stimulating discriminant measures (SDM) to achieve segmentation of the WBC. SDM measures were used with a GA to obtain clustering of nucleus, cytoplasm, and background regions. Rawat *et al.* [30] proposed a technique to segment the WBC nucleus using morphological operations and thresholding techniques. Image fusion using the principal component analysis was performed to achieve better segmentation results. Madhloom *et al.* [31] proposed a technique to localize and segment the nuclei using the arithmetic method and automatic threshold. Hegde *et al.* [32] proposed a technique to detect the region of nuclei using Rosin's thresholding method and histogram-based thresholding method. Rawat *et al.* [33] cropped the blood smear image to obtain a subimage consisting of the nucleus. Global Otsu's threshold and morphological operations were incorporated to obtain the final nucleus. A set of 331 features were extracted to classify the cells. The classification was done using the GA-SVM. Agaian *et al.* [34] designed a system to diagnose acute myelogenous leukemia. The nuclei were extracted using the K-means algorithm. Shape, texture, and color features, along with the Hausdorff dimension, were extracted to differentiate a noncancerous cell from a cancerous one. SVM was used as the classifier. Kazemi *et al.* [1] proposed a model to diagnose AML using the K-means clustering algorithm. The AML subtypes were classified using the multi-SVM. Different shape, texture, and color features were extracted to classify AML into its subtypes. Goutham *et al.* [35] developed a model to diagnose AML using the K-means clustering, local directional path, and SVM algorithm. Su *et al.* [36] developed a model to identify AML by using the K-means clustering algorithm and hidden-Markov random field (HMRF) segmentation method. The model was able to differentiate between six groups of cells and hence it helped in the determination of blast counting. Tran *et al.* [37] designed an automated method to segment the nucleus and cytoplasm to detect the AML and its subtypes. Otsu's threshold was used to obtain the nuclei, and the gradient of the image was computed to obtain the cytoplasm. Dasariraju *et al.* [38] employed multi-Otsu thresholding to segment nucleus and cytoplasm. Sixteen features were extracted and a random forest algorithm was employed to achieve the classification of the immature leukocytes.

Agus *et al.* [39] proposed a technique to classify AML to its subtypes using active contour without edge segmentation and momentum back-propagation artificial neural network. Six features were used for training the model. Markiewicz [40] proposed a system to diagnose acute myelogenous leukemia. The cells were extracted from the bone marrow aspirate using the watershed algorithm. Texture, geometrical, and statistical features were computed to achieve the classification of the cells. Priya *et al.* [41] proposed a technique to diagnose AML by extracting the nuclei using the Fuzzy C-Means clustering algorithm. The features were extracted, which were then fed to neural networks to achieve the classification. Matek [42] used ResNeXt CNN for the classification of images having different types of cells present in the blood smear. The model was trained for 20 epochs. Das *et al.* [43] used color space transformation technique to differentiate NRBCs from WBCs. The nucleated cell was extracted using the special fuzzy c-means clustering algorithm. It later differentiated NRBC from WBC based on different features extracted.

Based on these works, it was identified that an efficient CAD tool for AML diagnosis must include: 1) Effective algorithm for localization of the blast cells in the blood smear image and extraction of all vital features of both the WBC nucleus and cytoplasm of all the cells without cropping. 2) Algorithm to identify adjacency and overlapping between cells in the smear, which is robust to staining variation and illumination conditions.

A. Research Contributions

The main contributions of the article are as follows:

1) Segmentation of the nucleus of WBC and NRBC with different shapes and deformities from the blood smear image with staining variation using the K-medoids algorithm. Extraction of discriminative features to differentiate between the NRBCs and WBCs.
2) An algorithm to accurately separate touching cells and also deal with overlapping between the cells.
3) A novel automated algorithm to segment the cytoplasm of all the WBCs without involving cropping of cells and manual intervention.
4) To achieve the diagnosis of AML and classify it to the respective subtype based on the count of each type of WBC.

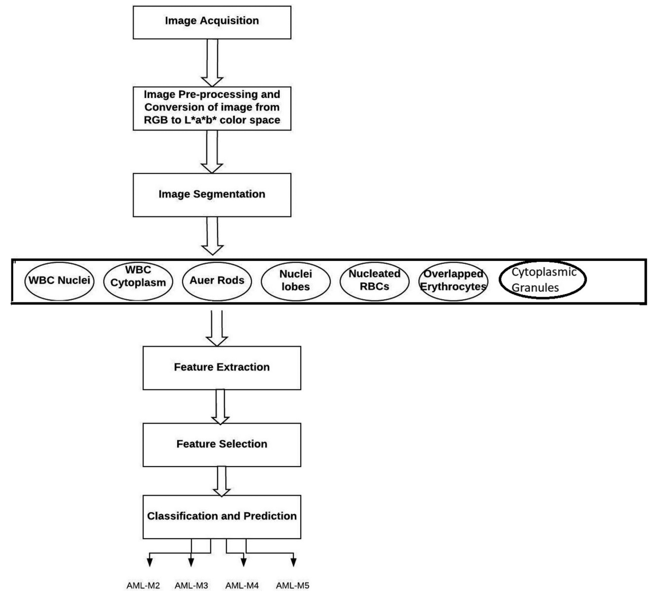


Fig. 2. Proposed model to achieve the diagnosis of AML.

The organization of the rest of the article is as follows. Section III discusses the datasets used and the proposed methodology, which includes algorithms for blood smear image segmentation. Section IV explains the results. Section V describes the comparative analysis, and Section VI concludes this article.

III. Materials and Methods

The development of the CAD to achieve the diagnosis and classification of acute myeloid leukemia generally consists of the following stages: Image acquisition, image preprocessing, image segmentation, feature extraction, feature selection, classification, and prediction. The steps involved in designing the CAD are shown in Fig. 2. MATLAB R2017b tool is used to implement the algorithms developed in this article.

A. Image Acquisition

1) To form the training set, 800 images consisting of AML M2–M5 samples were obtained from the Kasturba Hospital, Manipal. An Olympus U-CMAD3 camera was mounted on an optical microscope with a magnification of 4 *100 to capture the images. The U-CMAD3 was a single CCD digital color camera with 5.24 megapixels. Out of the 800, 400 images were stained using Leishman stain, and the other 400 using Wright–Giemsa stain. The May Grünwald–Giemsa stained 200 NRBC samples were obtained from the dataset of microscopic peripheral blood cell (PBC) images for the development of automatic recognition systems [44]. The format was JPEG, and the size of the images was 360*363 pixels. The resolution was increased to 400 *400. The training dataset consisted of 200 images under each category (M2, M3, M4, M5, and NRBC) that formed a total of 1000 images. All the images in the training dataset have a size of 400 *400 pixels, and JPEG is the image format.

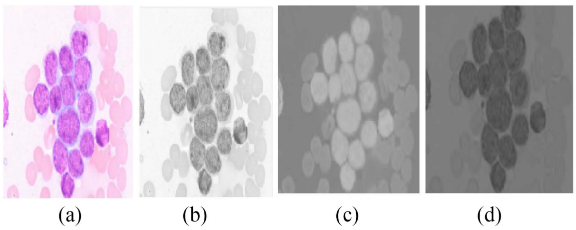


Fig. 3. (a) Original image. (b) L* component. (c) a* component. (d) b* component.

2) The test dataset consisted of 400 images consisting of AML M2–M5 samples which were obtained from the Kasturba Hospital, Manipal. Out of the 400, 200 images were stained using Leishman stain, and the other 200 using Wright–Giemsa stain. The May Grünwald–Giemsa stained 100 blood smear images having NRBCs were obtained from the PBC dataset. The images in the test set were separate from the training set. It consisted of 100 images under each category (M2, M3, M4, M5, and NRBC) that formed a total of 500 images. The expert provided annotation of cell types. The images have a size of 400*400 pixels and JPEG is the image format.

B. Image Preprocessing

The images generated by the digital microscope are in red green blue (RGB) format, making it difficult to segment. The image is converted from RGB to L*a*b* color space to achieve accurate segmentation by overcoming the issues related to staining variations. Color information is present in a* and b* layers [45], [46], and hence it is used for further segmentation. Fig. 3 shows the input image and the components of L*a*b* color space after its conversion.

C. Image Segmentation

The main aim of the segmentation stage is to separate the WBC and NRBC from the other surrounding components of blood, such as RBCs (erythrocytes), platelets, and blood plasma. Next, each cell is segmented into the nucleus and the cytoplasm.

1) Detection of Nuclei: The features of the nuclei play a vital role in the diagnosis of the disease. The detailed steps to obtain the WBC/NRBC (if present) nuclei are given in the algorithm 1. The blood smear image is segmented using the K-medoids algorithm. The color image is divided into clusters based on the distance criterion. K-medoids has the upper hand over the K-means algorithm as it is less sensitive to the outliers [47]. It operates by choosing the medoid of the objects in the cluster as the reference point as explained in [48]. The objective function of the K-Medoids algorithm is given in (1). Euclidean distance is the distance criterion, and the number of clusters is set to four. One of the clusters will contain nuclei, Auer rods (if present), and platelets (if present), and the other three clusters will contain cytoplasm, RBCs, and background. Since the smears are stained using Wright–Giemsa, Leishman stain, and May Grünwald–Giemsa stain, the WBC nucleus is colored

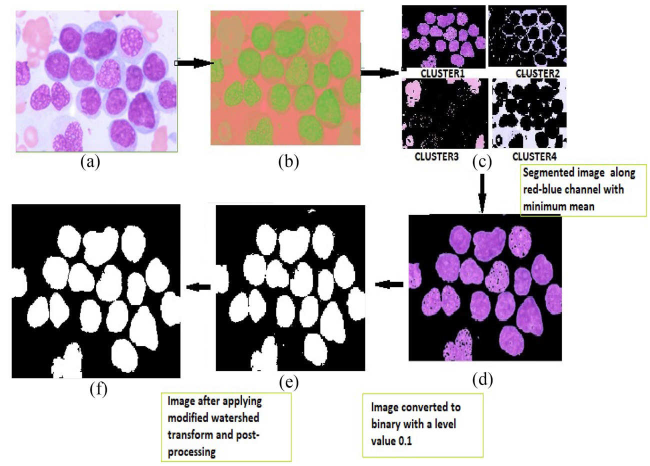


Fig. 4. (a) Original image. (b) Image converted to L*a*b* color space. (c) Four clusters having the segmented images after applying K-medoids. (d) Segmented image having the WBC nuclei. (e) Binary image of the WBC nuclei. (f) WBC nuclei after the application of modified watershed transform and postprocessing.

purple to violet. Hence, the segmented image along the red-blue channel highlights the nucleus. The block diagram that depicts the extraction of WBC nuclei is shown in Fig. 4. The mean value of the segmented images along the red-blue channel is tabulated in the Table I

$$V = \sum_{i}^{k} \sum_{x_j \epsilon S_i} (x_j - c_i)^2 \tag{1}$$

where k is number of clusters, S_i is a cluster, c_i is the medoid of S_i.

2) Modified Watershed Transform to Separate Touching Cells: The raw watershed transform considers each tiny local minima as the catchment basin. The proposed work utilizes the modified watershed transform to eliminate the drawback of oversegmentation. The tiny local minima are eliminated using the imextendedmin [51] function. The function produces dots at the center of every cell. The minima imposition using the imimposemin [52] function is used to modify the distance transform so that no minima occur at the filtered out location.

3) Identification of NRBCs and Its Differentiation From the WBCs: A nucleated red blood cell (erythroblast) contains a nucleus. Its presence indicates an underlying severe disease. NRBCs detection and quantification are still based on the microscopic analysis of stained blood. They are counted as WBCs by most of the automated hematology analyzers because of the presence of the retained nucleus. Due to this, it leads to an incorrect count of WBCs, which generates misleading results. Identification of NRBCs is vital to prevent incorrect count of WBCs [53].

In the proposed work features generated using the first-order statistics and the contrast of the nucleus of NRBC are mainly used to identify the NRBCs and differentiate them from the other WBCs. Both the cells have individual dissimilarity while considering the morphology and intensity information. The authors identified that NRBCs stain darker than the WBCs,

Algorithm 1: Extraction of Nuclei.

1: **procedure** Find_NUCLEI
2: Input: Blood smear image I(x,y) after conversion to L*a*b* color space, Output: Image N(x,y) with nuclei of cells.
3: Apply the K-medoids algorithm with four clusters and Euclidean distance as the distance metric. It classifies the color in a*b* space.
4: Label the pixels in the image with the result generated from K-medoids algorithm.
5: Obtain all the segmented images Q1(x,y),Q2(x,y),Q3(x,y), and Q4(x,y) of I(x,y) based on color

$$Q1(x,y) = \{R1(x,y), G1(x,y), B1(x,y)\} \quad (2)$$

$$Q2(x,y) = \{R2(x,y), G2(x,y), B2(x,y)\} \quad (3)$$

$$Q3(x,y) = \{R3(x,y), G3(x,y), B3(x,y)\} \quad (4)$$

$$Q4(x,y) = \{R4(x,y), G4(x,y), B4(x,y)\}. \quad (5)$$

6: Extract the red-blue channel from each of the segmented image as nuclei color will be higher in contrast. Obtain the mean of the images

$$M1 = \text{mean}\{R1(x,y), B1(x,y)\} \quad (6)$$

$$M2 = \text{mean}\{R2(x,y), B2(x,y)\} \quad (7)$$

$$M3 = \text{mean}\{R3(x,y), B3(x,y)\} \quad (8)$$

$$M4 = \text{mean}\{R4(x,y), B4(x,y)\}. \quad (9)$$

7: Choose the segmented image with minimum mean to obtain the image containing nuclei N_M(x,y).
8: The obtained nuclei are converted to binary using the below binary conversion function [49] with a level value 0.1 (chosen after experimental analysis)

$$B(x,y) = im2bw(N_M(x,y), 0.1).$$

9: Eliminate platelets by removing all objects in *B*(*x*,*y*) that are less than 200 pixels.
10: Separate the touching nuclei using the modified watershed transform to obtain final image *N*(*x*,*y*).
11: Label the nuclei and count them using the bwlabel [50] function

$$[LabeledImage, number\_of\_Objects] = bwlabel(N(x,y)).$$

12: **end procedure**

and the WBC nucleus was higher in contrast than the NRBC nucleus. Hence, the computation of the intensity information of the nucleus played an essential role in distinguishing the two types of cells. The (10) computes the first-order histogram (H). ‘I’ represents the pixel intensities in the region of interest (ROI), which is the nucleus of WBC and NRBC, and “B” represents the equally spaced bins. In the proposed work, a default value of 256 bins is used

$$H(i) = \frac{\text{Number of pixels with gray levels in } I\epsilon B_i}{\text{Number of pixels in the image}}. \quad (10)$$

Based on the definition of H(I), the mean m_1 and central moments μ_k of the image “I” (grayscale scale image having WBC/NRBC nuclei) is calculated using the (11) and (12). The variance is measured by computing the width of the histogram that depicts the deviation of the gray levels from the mean. The sharpness of the histogram represents kurtosis. The difference between the highest and lowest intensity value of the image represents the contrast

$$m_1 = \sum_{I=0}^{N_L-1} I^1 H(I) \quad (11)$$

$$\mu_k = \sum_{I=0}^{N_L-1} (I - m_1)^k H(I) \quad (12)$$

where k = 2, 3, 4 and N_L represents number of gray levels
μ_2 gives variance
μ_3 gives skewness
μ_4 gives kurtosis

4) Extraction of the Cytoplasm: The features of cytoplasm play a significant role in identifying the stage of the AML [54]. The nucleus-cytoplasm ratio (N/C) is a vital feature that helps to differentiate a malignant cell from a nonmalignant one [55]. A higher N/C ratio indicates malignancy. It is identified that the normal cells always have an N/C ratio below 0.5, and malignant cells have an N/C ratio above 0.5. It also plays a major role in differentiating the various types of abnormal WBCs in the blood. The detailed steps to extract the cytoplasm of all the cells are given in the algorithm 2. Fig. 5 depicts a flowchart consisting of steps to extract the cytoplasm of the cell. The size of WBCs is larger than the erythrocytes. Hence, in the proposed work, they are separated from surrounding erythrocytes by utilizing their size details. In some cases, as a result of overlapping, the size of the erythrocytes matches the size of the WBCs. Due to this, they are misclassified as WBCs. In the proposed work, a novel technique to prevent misclassification is presented. Since the WBC has nuclei and erythrocytes do not have nuclei, the Euclidean distance between the centroid of WBC nuclei and the object identified as the WBC is computed. A threshold value of 12 pixels is chosen (after experimental analysis). It is noticed that since the erythrocytes do not possess a nucleus, the Euclidean distance between its centroid and centroid of the detected WBC nucleus is always greater than 12 pixels.

5) Extraction of the Auer Rods: Auer rods are clumps of granular materials seen in the cytoplasm of the blast cells. They are found in M2 and M3 subtypes of AML. In the proposed work, a technique is developed to diagnose them. The rods are stained with the red color that can be captured using the segmented image with minimum mean along the red-blue channel. Due to its needle-like shape, it can easily be differentiated from the WBC nuclei using the form factor (FF) computed using (13). The area(A) represents the actual number of pixels in the region. The perimeter(P) gives the distance around the boundary of

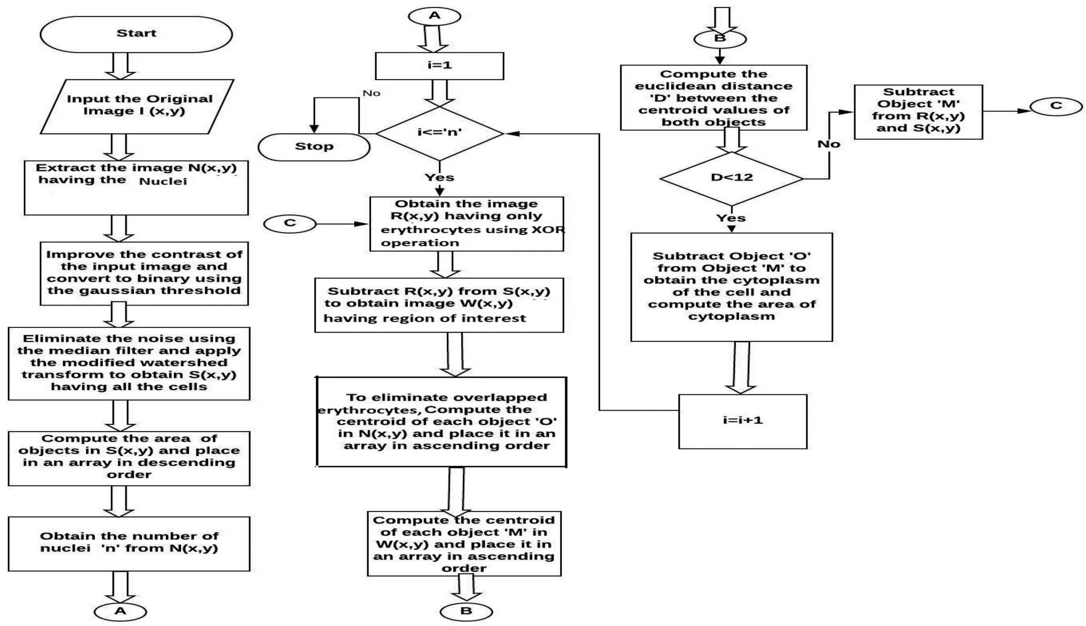


Fig. 5. Flowchart to depict the extraction of cytoplasm.

the region. Since the shape of Auer rods (needle-like shape) deviates from a circular shape, the value of 0.6 used as FF helps in identifying the Auer rods. The values are chosen after the experimental analysis of the images present in the training dataset. The form factor computation helps to also differentiate between platelets and Auer rods. Auer rods are smaller compared to the WBC nuclei; hence they have a smaller area than the nuclei. A bounding box is drawn around the Auer rods and the corresponding cytoplasm of the cell to verify its presence in the cytoplasm and eliminate the possibility of selection of other artifacts as Auer rods. The coordinates are then compared to ensure its position in the cytoplasm

$$\text{Form\_factor}(FF) = 4 * pi * A/(P * P). \tag{13}$$

6) Detection of Multiple Lobes in the WBC Nuclei: The promyelocytes are responsible for acute promyelocytic leukemia, a subtype of AML (AML-M3). The cytoplasm of the cell has granules that are stained red to purple and the nuclei of the cell are bilobed to multiple lobed [59]. The presence of numerous lobes in nuclei and primary granules in the cytoplasm helps to differentiate the atypical promyelocyte from the other cells (myeloblast, promonocyte, monoblast, and NRBC) [60]. Furthermore, they can be differentiated from the neutrophils and eosinophils based on the nuclear chromatin pattern and the presence of the primary granules in the cytoplasm. According to the domain experts, the nucleus to cytoplasm area ratio (N/C ratio) of promyelocytes will always be less than or equal to 0.5 due to the presence of nuclei. To capture the primary granules, we compute the texture features of the cytoplasm. Even though neutrophils and eosinophils are myeloid cells, they can be differentiated from each other based on their structure. The diameter of the eosinophil is larger than the diameter of the neutrophil [61]. In the proposed work, this diameter is computed by calculating the perimeter of the WBC nuclei. The detailed steps to detect the presence of multiple lobes in nuclei of each cell and granules are given in the algorithm 4 and algorithm 5.

D. Evaluation Metrics to Measure Segmentation Accuracy

The efficacy of the proposed system in segmentation is evaluated using the structural similarity index (SSIM), boundary F1 (BF Score), dice coefficient (DC), Jaccard coefficient (JC), precision (P_c), recall (R_c), and accuracy [62]. The binary form of the images is taken into consideration. SSIM is computed using

$$\begin{aligned}\text{SSIM}(x, y) = (2\mu_x\mu_y + c1)(2\sigma_{xy} + c2)/ \\ (\mu_x^2 + \mu_y 2 + c1)(\sigma_x^2 + \sigma_y 2 + c2)\end{aligned} \tag{14}$$

where *x* represents the segmented image, y represents the ground truth, μ_x and μ_y represent the means of segmented image and ground truth image, σ_{xy} represent the covariance, c1 and c2 are the constants to deal with situations when the denominator value is close to zero.

The BF score (BF) is computed using the (15)

$$BF = 1/k_{cl}. \sum (2 * P_c * R_c)/(P_c + R_c) \tag{15}$$

where k_{cl} is number of different classes, P_c is precision and R_c is recall.

E. Features Extraction and Feature Selection

The main aim of this step is to obtain the descriptors that match the visual patterns that the pathologists use to identify the disease. The shape features [63] are extracted to measure the geometrical appearances of blast cells that agree with the domain expert's discernment. Some of the features like eccentricity and

Algorithm 2: Extraction of the Cytoplasm of Cells and Computation of Area.

1: **procedure** Cytoplasm Extraction
2: Input: Original blood smear image, Output: Cytoplasm area
3: Extract the nucleus *N*(*x*,*y*) from the image *I*(*x*,*y*) using the steps described under the Section III-C1.
4: Improvise the contrast of the image *I*(*x*,*y*) by increasing the local contrast of the image using the fast local Laplacian filter function with $a_m = 0.4$ and $S_r = 0.1$ and convert it to grayscale.
5: Convert the image to binary using the Gaussian Otsu's threshold [56].
6: Median filter is applied with 3*3 neighborhood to eliminate noise by replacing the pixel by the median of pixels around neighborhood

$$y[p,q] = \text{median} x[i,j], (i,j)\epsilon w$$

w denotes the neighborhood centered around the location (p,q).
7: Apply the modified watershed transform to separate the touching objects to obtain the image *S*(*x*,*y*).
8: Calculate the area of the objects "O" present in the image *S*(*x*,*y*)

$$\text{Area}_o = |p(x,y)|p(x,y)\epsilon O| \,.$$

9: Place the computed value in an array "Q" in decreasing order.
10: Obtain the number of nuclei "*n*" from the extraction step.
11: **for** $1 : n$ **do**
12: Obtain the image *R*(*x*,*y*) having only RBCs using the size based analysis. To keep objects between lower bound [area in pixels] and upper bound [area in pixels], the bwareaopen function can be used with XOR

$$R(x,y) = \text{XOR}(\text{areaopen}(S(x,y), m1),$$
$$\text{areaopen}(S(x,y), m2))$$

where $m1$=1 is lower bound and $m2$=Q(n) is upper bound XOR: gives the absolute difference between two images.
13: **end**
14: Subtract *R*(*x*,*y*) from *S*(*x*,*y*) to get image *W*(*x*,*y*) with only WBCs and erythroblasts (if present).
15: **for** each object "O" ϵ *N*(*x*,*y*)**do**
16: Compute the centroid of "O" and place it in ascending order in an array
17: **end**
18: **for** *each object "M" ϵ W(x,y)***do**
19: Compute the centroid of "M" and place it in ascending order in an array.
20: **end**
21: Compute the Euclidean distance "D" between centroid values of object "O" and object "M";

$$D = \sqrt{(x2-x1)^2, (y2-y1)^2}.$$

22: **if** $D < 12$ **then**
23: Subtract Object 'O' from Object 'M' to obtain the cytoplasm and Compute the area of the cytoplasm using the equation mentioned in step 8.
24: **else**
25: Subtract Object 'M' from image S(x,y) and R(x,y) and goto step 12 and repeat the procedure.
26: **end**
27: **end procedure**

Algorithm 3: Extraction of Auer Rods.

1: **procedure** $Find\_Auer\_Rods$
2: Input: Segmented image $N_M(x,y)$ with minimum mean along the red-blue channel, Output: Auer rods.
3: The obtained image is converted to binary *B*(*x*,*y*) with a level value 0.1 (chosen after experimental analysis)

$$B(x,y) = im2bw(N_M(x,y), 0.1)).$$

4: Eliminate the platelets and other artifacts using the below areaopen function [57]

$$B_M(x,y) = \text{bwareaopen}(B(x,y), 20).$$

5: Separate the touching cells using the modified watershed transform.
6: Obtain the area of objects that have a form factor greater than 0.6. Store it in an array "A" in ascending order

$$\text{Minimum\_Area} = A[0].$$

7: Compute the area of all objects present in the resulting image.
8: Find the objects in $B_M(x,y)$ which have a form factor less than 0.6 and area less than the minimum area (in pixels) using the below find function [58]

$$\text{Rods} = \text{find\_}B_M(x,y)(FF < 0.6 \text{ and Area}$$
$$< \text{Min\_Area}).$$

9: Draw a bounding box around the Auer rods and obtain its coordinates

$$\text{Bounding\_Box\_1} = (x, y, \text{width}, \text{height}).$$

10: Draw a bounding box around the WBC cytoplasm and obtain its coordinates

$$\text{Bounding\_Box\_2} = (px1, px2, ax1, ax2)$$

px1 = xLeft, px2 = yTop, ax1 = width, and ax2 = height
11: **if** *(px1 <= x && x <= ax1 && px2 <= y && y <= ax2)***then**
12: Return "Yes"
13: **else**
14: Return "No"
15: **end**
16: **end procedure**

Algorithm 4: Detection of Granules in the Cytoplasm.

1: **procedure** Detect Granules
2: Input: Original image and binary image of cytoplasm obtained from Section III-C4, Output: Granules in the cytoplasm
3: Use the binary image of the cytoplasm as the mask
4: In the original image, set the pixels outside the mask region to zero
5: Convert the resulting image to gray scale
6: Compute the texture features of the resulting image to detect the granules present in the cytoplasm
7: **end procedure**

Algorithm 5: Detection of Multilobed Nuclei.

1: Input: Segmented image $N_M(x, y)$ and $W(x,y)$ having WBCs and erythroblasts (if present), Output: Multilobed or bilobed nuclei.
2: **for** *each object* "*O*" in $W(x,y)$ **do**
3: Obtain the largest "x," largest "y," smallest "x," smallest "y" points of the blob

$$[r1, c1] = \text{find}(\text{img} == \min(\min(\text{img})))$$

$$[r2, c2] = \text{find}(\text{img} == \max(\max(\text{img})))$$

where r1,c1,r2, and c2 are row and column of the image and img is $W(x,y)$.
4: Utilize the above coordinates to obtain subimage *S_N*(*x*,*y*) of the nuclei belonging to the corresponding cell.
5: The obtained nuclei are converted to binary using level value 0.1 and modified watershed transform is applied. The number of regions in the image is stored in "C.
6: **end**
7: **if** C $>$ 1**then**
8: Return "Yes"
9: **else**
10: Return "No"
11: **end**

solidity provided the semantic meaning of the blast cell, which is identical to the domain expert's knowledge. Histogram-based features [64] and the GLCM features [65] are the two types of texture features extracted. It helps to capture the nuclear chromatin pattern. Values of GLCM parameters depend much more on textural patterns of the interior of the nuclei than on the precision of nuclear boundary determination [66]. Nuclear texture analysis helps in the assessment of interchromosomal coarseness [67]. The grayscale image of the nucleus is utilized here. The intensity information played an important role in distinguishing the WBCs from erythroblasts. The chromatin pattern played a vital role in identifying the different types of cells (myeloblast, promonocyte, monoblast, myelocyte, promyelocyte, neutrophil, and eosinophil). Additionally, the difference

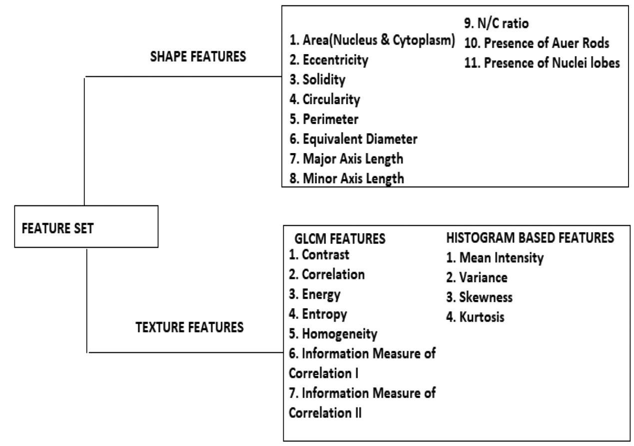


Fig. 6. Feature set.

in the nuclear chromatin pattern of myeloblasts helped to differentiate them from neutrophils and eosinophils. The nuclear chromatin pattern of myeloblasts is fine, whereas the nuclear chromatin pattern of neutrophils and eosinophils is coarse and clumpy. The features extracted are shown in Fig. 6. The irrelevant features are eliminated by applying feature selection to improve the accuracy of the algorithm. Feature selection is achieved using the information gain. Information gain measures the extent to which each feature gave information about the class. It is a measure of the reduction in the entropy, where entropy represented the impurity in the collected sample. The entropy is computed using the (16). The gain is calculated using the (17)

$$\text{Entropy} = -\sum_{i}^{C} p_i * \log_2 p_i \tag{16}$$

where p_i probability with which element of class *i* is selected. C denotes the number of classes

$$\text{Gain}(A, B) = H(A) - \sum_{V \epsilon \text{Values}(B)} (|A_v| \,/\, |A|) * H(A_v) \tag{17}$$

where *B* denotes attribute, *A* represents the sample X, *V* denotes the possible values of an attribute, A_v represents the subset, where $X_B = V$.

InfoGainAttributeEval in Waikato environment for knowledge analysis (WEKA) tool [68] gave the information gain. The ranker search method gave the ranks to the attributes. The feature indicating the presence of Auer rods is set to "Yes" or one when the rods are detected or set to "No" or zero when not detected. The feature indicating the presence of nuclei lobes is set to "Yes" or one if nuclei have lobes or set to "No" or 0 if lobes are not present.

F. Counting of the Cells

The AML subtype is determined based on the percentage count of each type of cell (myeloblast (A), promonocyte (B), monoblast (C), myelocyte (D), promyelocyte (E), eosinophil (F), neutrophil (G), and NRBC (H)). Table II demonstrates the

TABLE I
MEAN VALUE OF THE SEGMENTED IMAGES ALONG THE RED-BLUE CHANNEL

Cluster 1	Cluster 2	Cluster 3	Cluster 4
0	0.6325	100.2125	142.6350

TABLE II
AML SUBTYPE BASED ON PERCENTAGE COUNT OF EACH TYPE OF CELL

Type	A	B	C	D	E	F	G	H
M2	30-89	<10	<20	<10	<10	<15	>10	<10
M3	<30	<10	<10	<10	>90	<10	<10	<10
M4	>30	>20	>20	>10	<10	>10	<10	<10
M5	<10	>10	>80	<10	<10	<10	<20	<10
Other	<10	<10	<10	<10	<10	>10	>10	>10

classification of AML to its subtypes based on the percentage of cells.

Equation (18) is used to compute the percentage of each type of cell in the smear.

$$\text{Percentage Count} = \text{Count of each type of cell}/\text{Total count}. \tag{18}$$

G. Classification and Prediction

In this research article, the classification algorithms are run in WEKA. The algorithms classify the WBC into different types (myeloblast, promyelocyte, promonocyte, myelocyte, monoblast, neutrophil, eosinophil, and NRBC) and compute the percentage of each cell to achieve the diagnosis of the stage of the AML. The work focuses on identifying the four main stages of AML, namely, M2/M3/M4/M5 and a stage "Other," when the count of NRBCs exceeds a range as specified in Table II. The stage "Other" may indicate diseases like severe anemia. Based on the percentage count of each type of cell, the final stage is determined. The different classifiers, namely, naive Bayes [69], decision tree [70], K-nearest neighbor (KNN) [71], and random forest [72], are compared. The algorithms are run in WEKA.

In the proposed article, a comparative analysis is done between the supervised classifiers. Image level is used as the benchmark for the classification. The performance of the random forest algorithm, decision tree, naive Bayes, and KNN is compared. The sensitivity, specificity, positive predictive value (PPV), negative predictive value (NPV), and accuracy are computed [73]. The average accuracy is computed using the (19)

$$\text{Average\_Accuracy} = \text{A1/T1} \tag{19}$$

where A1 : Accuracy in detecting each subtype T1: Total number of subtype

IV. RESULTS

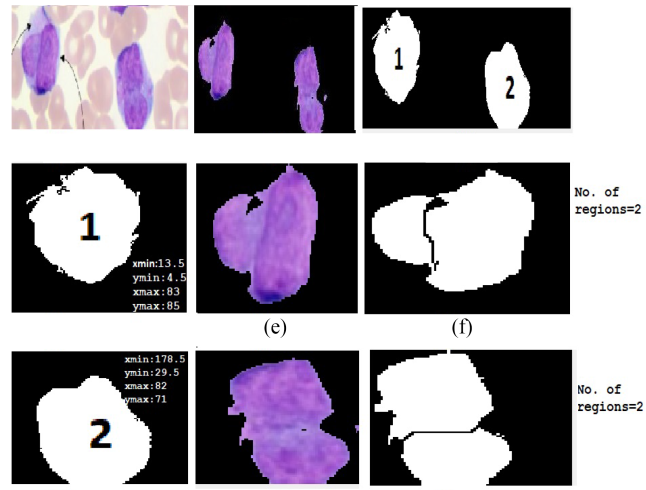
The developed CAD system's performance to diagnose AML is assessed with the use of blood smear images stained with different stains and having cells with varying deformities. The gold reference standard is obtained through the interpretation of the image by the pathologist.

A. Result of Extraction of WBC Nucleus and Cytoplasm

The average values of segmentation performance measures for all the images in the training and test data are reported under Section IV-C. Fig. 7 shows the detection of bilobed nuclei in an AML-M3 sample, as explained in Section III-C6. Fig. 8 shows the extraction of granules present in the cytoplasm of a promyelocyte cell.

Fig. 9 illustrates the result of the extraction of WBC nuclei and cytoplasm from some sample images present in training set by using the algorithm explained in Section III-C1 and III-C4.

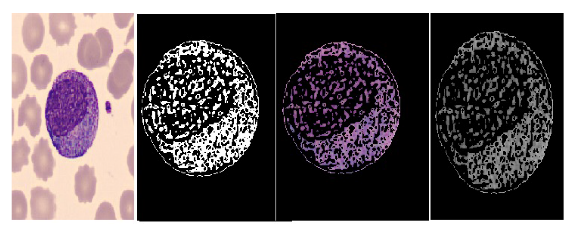
Fig. 10 illustrates a case where the overlapped erythrocyte is counted as WBC due to its size. Table III shows the centroid values of WBC nuclei and objects (erythrocyte) identified as WBC. It also shows the Euclidean distance between them. It is seen that the Euclidean distance between the centroid of the



(e) (f)

(g) (h) (i)

Fig. 7. (a). Original image. (b) Image $N_M(x,y)$ consisting the nuclei. (c) Image $W(x,y)$ having the labeled WBC. (d) First WBC with its coordinate value. (e) Corresponding nuclei extracted using the coordinate value. (f) Nuclei detected with lobes after applying the modified watershed transform. (g) Second WBC with its coordinate value. (h) Corresponding nuclei extracted using the coordinate value. (i) Nuclei detected with lobes after applying the modified watershed transform.



(a) (b) (c) (d)

Fig. 8. Illustration of the detection of granules present in the cytoplasm. (a) Original image having the promyelocyte. (a) Binary image representing the cytoplasm of the cell (mask). (c) Cytoplasm extracted from the original image after applying the mask. (d) Gray scale image of the cytoplasm with granules.

TABLE III
EUCLIDEAN DISTANCE (IN PIXELS) BETWEEN THE CENTROIDS OF WBC NUCLEI AND WBC

Object Label	Centroid of Nucleus	Centroid of WBC	Distance
1	(59.48,120.57)	(207.74,225.221)	181.47
2	(256.70,98.29)	(258.37,100.275)	2.5941
3	(304.67,217.58)	(306.562,207.64)	10.1096
4	(385.34,138.314)	(384.50,138.02)	0.8887

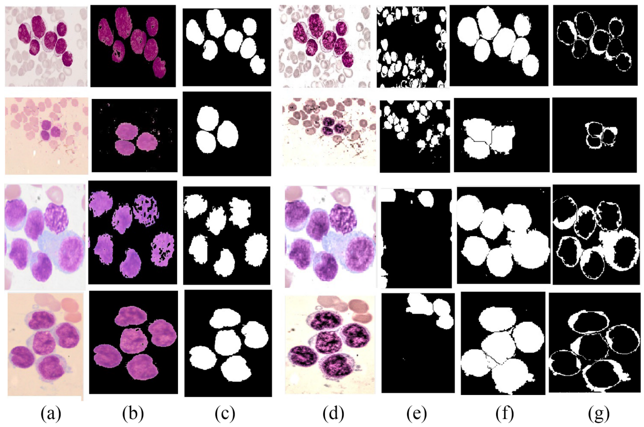


Fig. 9. Illustration of WBC segmentation. (a) Original image. (b) WBC nuclei extracted from the segmented image with minimum mean along the red-blue channel. (c) Binary image of the nuclei (after applying the modified watershed transform). (d) Image after contrast improvement. (e) Binary image of red blood cells. (f) Binary image of WBC cells. (g) Binary image of WBC cytoplasm.

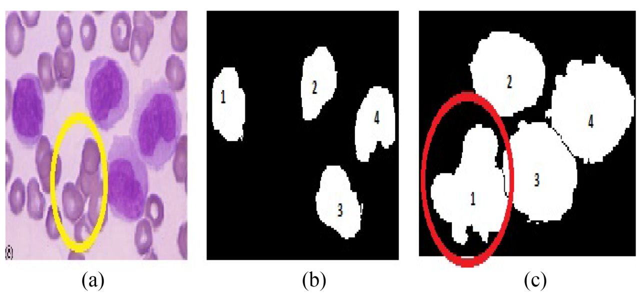


Fig. 10. (a) Original image with overlapping erythrocyte marked in yellow color. (b) Binary image having the WBC nuclei. (c) Overlapping erythrocyte misinterpreted as the WBC.

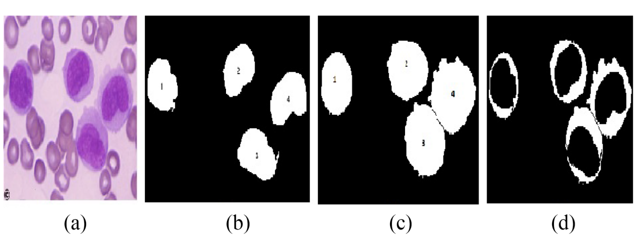


Fig. 11. (a) Original image with overlapping erythrocyte. (b) Binary image having the WBC nuclei. (c) WBC after eliminating the overlapping erythrocyte. (d) WBC Cytoplasm.

overlapped erythrocyte and the corresponding WBC nucleus is greater than 12. Hence, it is eliminated from consideration, as explained in Section III-C4. Fig. 11 shows the result after the elimination of erythrocyte. The centroid values and Euclidean distance between the centroid of the WBC nuclei and the actual cells (true positive) are tabulated in Table IV.

B. Result of the Extraction of the Auer Rods

Fig. 12 illustrates the extraction of the Auer rods from some of the sample images present in the training dataset. Table V

TABLE IV
EUCLIDEAN DISTANCE (IN PIXELS) BETWEEN THE CENTROIDS OF WBC NUCLEI AND WBC

Object Label	Centroid of Nucleus	Centroid of WBC	Distance
1	(59.48, 120.57)	(62.049, 121.901)	2.8933
2	(256.70, 98.29)	(258.37, 100.275)	2.5941
3	(304.67, 217.58)	(306.562, 207.64)	10.1096
4	(385.34, 138.314)	(384.50, 138.02)	0.8887

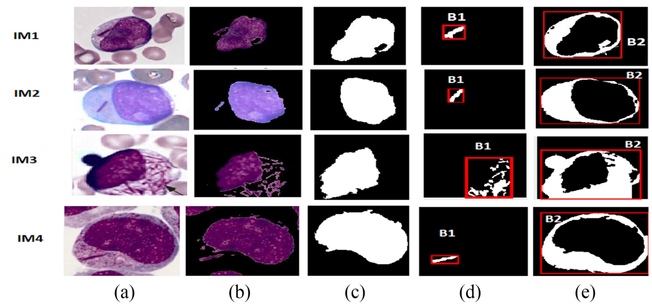


Fig. 12. Illustration of extraction of Auer rods. (a) Blood smear image. (b) WBC Nuclei with Auer rods. (c) Binary image of the WBC nucleus. (d) Bounding box drawn around the Auer rod. (e) Bounding box drawn around the WBC cytoplasm.

TABLE V
VERIFICATION OF THE PRESENCE OF AUER RODS WITHIN THE CYTOPLASM

Image	Coordinates of B1	Coordinates of B2	Auer Rods
IM1	(46.5,67.5,25,25)	(32.5,27.5,121,115)	Yes
IM2	(48.5,116.5, 19,35)	(18.5,38.5,183,164)	Yes
IM3	(58.5,52.5,59,77)	(5.5,30.5,122,100)	Yes
IM4	(12.5,13.5,173,137)	(24.5,108.5,42,17)	Yes

TABLE VI
SEGMENTATION PERFORMANCE PARAMETERS FOR THE TRAINING DATASET

Region	SSIM	BF Score	DC	JC	P_C	R_C	Accuracy
WBC nucleus	95	90	96	98	97	99	98.68
WBC	95	93	90	97	95	90	98.65
NRBC	98.8	99	97.1	94.42	99.67	94.74	99.81
Auer Rods	97.63	92	96.2	95.12	98	96.01	99.5

TABLE VII
SEGMENTATION PERFORMANCE PARAMETERS FOR THE TEST DATASET

Region	SSIM	BF Score	DC	JC	P_C	R_C	Accuracy
WBC nucleus	95	90	94	99	91	95	99
White blood cell	96	95	97	99	96	97	99.2
NRBC	99.01	99	99.2	98.43	98.53	99.9	99.86
Auer Rods	99.14	99.67	98.81	98.85	98.80	96.6	99.87

tabulates the coordinate values of the bounding box drawn around the Auer rods and cytoplasm to ensure its presence inside the cytoplasm, as explained in Section III-C5.

C. Evaluation Metrics

The average values of segmentation performance parameter values (in percent) for WBC nucleus, cell, NRBC, and Auer rods are tabulated in Tables VI and VII.

D. Result of Feature Selection

The features obtained after applying the feature selection are tabulated along with information gain in Table VIII.

TABLE VIII
RANKED ATTRIBUTES

Information Gain	Attribute	Information Gain	Attribute
2.0766	Carea	1.103	Major Axis$_{Nucleus}$
1.6914	Skewness$_{Nucleus}$	0.9601	Entropy$_{Nucleus}$
1.6234	Perimeter$_{Nucleus}$	0.8709	Mean Intensity$_{Nucleus}$
1.5913	Homogeneity$_{Nucleus}$	0.8709	Energy$_{Cytoplasm}$
1.3266	Minor Axis$_{Nucleus}$	0.8709	Kurtosis$_{Nucleus}$
1.3085	N/C ratio	0.8709	Contrast$_{Nucleus}$
1.1898	Eccentricity$_{Nucleus}$	0.835	Variance$_{Nucleus}$
1.1414	Narea	0.77935	Auer Rods
0.7076	Solidity$_{Nucleus}$	0.5138	Lobes$_{Nucleus}$
0.5	Energy$_{Nucleus}$		

TABLE IX
MODEL PARAMETERS

Algorithm	Parameters	Confusion Matrix					
		Matrix	M2	M3	M4	M5	Other
Random Forest	Features to Split= 4 ie:sqrt(Numb Number of Trees=100 Max_Depth= None	M2	198	0	02	0	0
		M3	02	197	01	0	0
		M4	0	04	195	01	0
		M5	0	0	02	198	0
		Other	0	0	01	0	199
		Matrix	M2	M3	M4	M5	Other
Decision Tree	Confidence Factor=25% Number of Folds=10 Max_Depth= -1(will be determin Minimum number of instances pe unpruned=False	M2	195	03	02	0	0
		M3	03	195	02	0	0
		M4	0	04	194	02	0
		M5	0	0	06	194	0
		Other	0	0	01	0	199
		Matrix	M2	M3	M4	M5	Other
Naive Bayes	Set the useKernelEstimator=True	M2	193	04	03	0	0
		M3	03	193	04	0	0
		M4	0	02	194	04	0
		M5	0	0	07	193	0
		Other	0	0	02	0	198
		Matrix	M2	M3	M4	M5	Other
KNN	Number of Neighbors =1 Distance Metric = Manhattan Search Algorithm= Linear NN Sc	M2	197	0	03	0	0
		M3	03	197	0	0	0
		M4	0	05	194	01	0
		M5	0	0	02	198	0
		Other	0	0	01	0	199

TABLE X
PERFORMANCE EVALUATION OF THE CLASSIFIERS

Metrics	Random Forest	Decision Tree	Naive Bayes	KNN
PPV$_{M2}$	99.00%	98.48%	98.47%	98.50%
Sensitivity$_{M2}$	99.00%	97.50%	96.50%	98.50%
Specificity$_{M2}$	99.75%	99.63%	99.63%	99.63%
NPV$_{M2}$	99.75%	99.38%	99.13%	99.63%
Accuracy$_{M2}$	99.60%	99.20%	99.00%	99.40%
PPV$_{M3}$	98.01%	96.53%	96.98%	97.52%
Sensitivity$_{M3}$	98.50%	97.50%	96.50%	98.50%
Specificity$_{M3}$	99.50%	99.13%	99.25%	99.38%
NPV$_{M3}$	99.62%	99.37%	99.13%	99.62%
Accuracy$_{M3}$	99.30%	98.80%	98.70%	99.20%
PPV$_{M4}$	97.01%	94.63%	92.38%	97.00%
Sensitivity$_{M4}$	97.50%	97.00%	97.00%	97.00%
Specificity$_{M4}$	99.25%	98.63%	98.00%	99.25%
NPV$_{M4}$	99.37%	99.25%	99.24%	99.25%
Accuracy$_{M4}$	98.90%	98.30%	97.80%	98.80%
PPV$_{M5}$	99.50%	98.98%	97.97%	99.50%
Sensitivity$_{M5}$	99.00%	97.00%	96.50%	99.00%
Specificity$_{M5}$	99.88%	99.75%	99.50%	99.88%
NPV$_{M5}$	99.75%	99.25%	99.13%	99.75%
Accuracy$_{M5}$	99.70%	99.20%	98.90%	99.70%
PPV$_{Other}$	100.00%	100.00%	100.00%	100.00%
Sensitivity$_{Other}$	99.50%	99.50%	99.00%	99.50%
Specificity$_{Other}$	100.00%	100.00%	100.00%	100.00%
NPV$_{Other}$	99.88%	99.88%	99.75%	99.88%
Accuracy$_{Other}$	99.90%	99.90%	99.80%	99.90%

E. Accuracy of the Classifier

The confusion matrix and model parameters of the classification algorithms are given in Table IX. The results of image-level classification are shown here. Count of the cells is used to classify the images into AML subtypes, as explained in Section III-F and III-G. The evaluation metrics are given in Table X. The performance of the classifiers is plotted in Fig. 13.

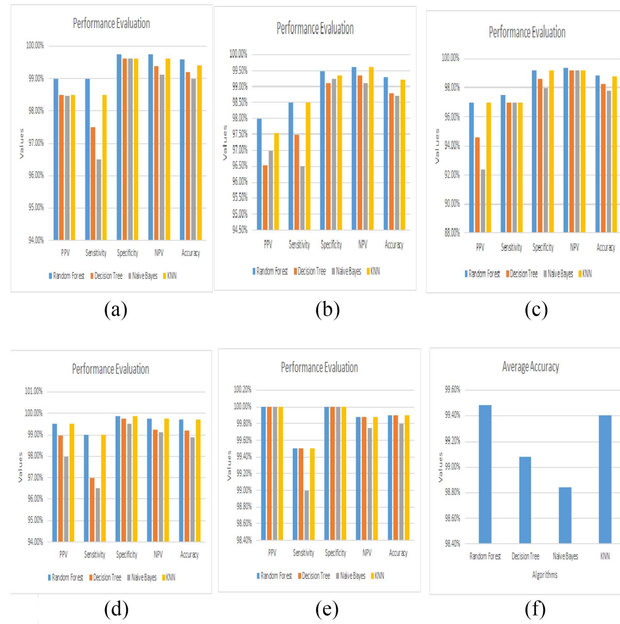


Fig. 13. (a) Performance of the classifiers in diagnosing AML-M2. (b) Performance of the classifiers in diagnosing AML-M3. (c) Performance of the classifiers in diagnosing AML-M4. (d) Performance of the classifiers in diagnosing AML-M5. (e) Performance of the classifiers in diagnosing other type (NRBC). (f) Average accuracy of the classifier.

TABLE XI
CONFUSION MATRIX OF TEST SET AND PERFORMANCE OF THE RANDOM FOREST ALGORITHM

Metrics	Random Forest				
Matrix		M2	M3	M4	M5	Other
	M2	97	0	03	0	0
	M3	0	98	02	0	0
	M4	0	02	98	0	0
	M5	0	0	01	99	0
	Other	0	0	02	0	98
Sensitivity$_{M2}$	97%					
Specificity$_{M2}$	100%					
PPV$_{M2}$	100%					
NPV$_{M2}$	99.25%					
Accuracy$_{M2}$	99.4%					
Sensitivity$_{M3}$	98%					
Specificity$_{M3}$	99.5%					
PPV$_{M3}$	98%					
NPV$_{M3}$	99.5%					
Accuracy$_{M3}$	99.2%					
Sensitivity$_{M4}$	98 %					
Specificity$_{M4}$	98%					
PPV$_{M4}$	92.45%					
NPV$_{M4}$	99.49%					
Accuracy$_{M4}$	98%					
Sensitivity$_{M5}$	99 %					
Specificity$_{M5}$	100%					
PPV$_{M5}$	100%					
NPV$_{M5}$	99.75%					
Accuracy$_{M5}$	99.8%					
Sensitivity$_{Other}$	98%					
Specificity$_{Other}$	100%					
PPV$_{Other}$	100%					
NPV$_{Other}$	99.5%					
Accuracy$_{Other}$	99.6%					

Random forest classifier outperformed the other classifiers, as shown in Table X. Hence, it was used for the prediction of the cell types present in the test set images. The confusion matrix of the test set and the random forest algorithm's performance as the classifier is tabulated in Table XI.

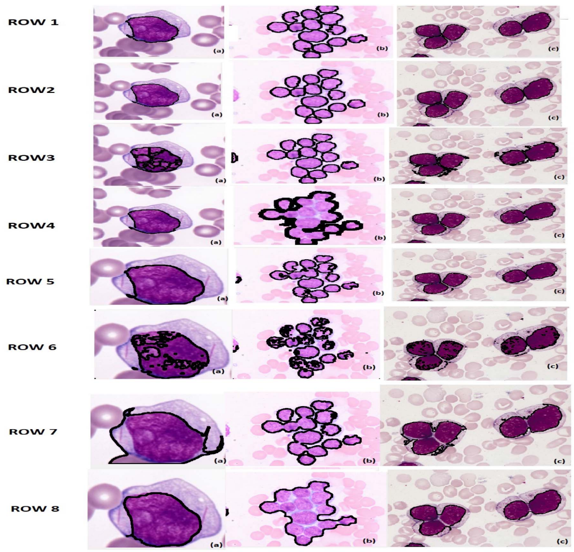


Fig. 14. Nuclei detection result; Row1 ground truth, Row 2 results of the proposed method, Row 3 results of technique in [22], Row 4 results of technique in [24], Row 5 results of technique in [30], Row 6 results of technique in [31], Row 7 results of technique in [32], Row 8 results of method in [20].

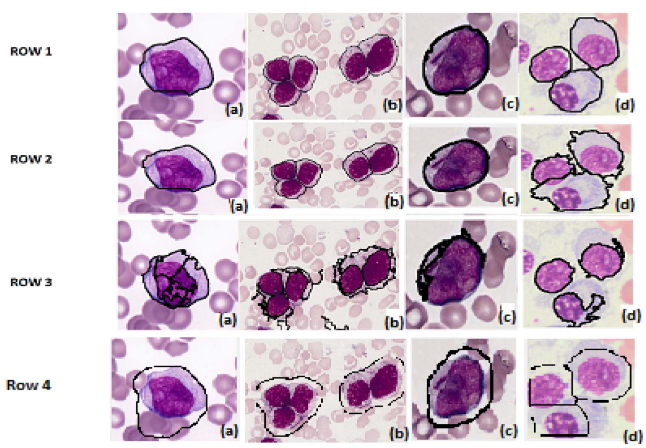


Fig. 15. Cell detection result; Row1 ground truth, Row 2 results of the proposed method, Row 3 results of the technique in [22], Row 4 results of the technique in [20].

V. Discussion

In this article, we implemented the segmentation of WBC (nucleus/cell) on our training dataset as per the method proposed in [20], [22], [24], [30]–[32]. The result of nuclei detection on few sample images (training set) achieved by the proposed method along with expert annotated images and results obtained using the technique in [20], [22], [24]–[32] is depicted in Fig. 14. The result of cell detection on few sample images (training set) achieved by the proposed method along with expert annotated images and results obtained using the technique in [20] and [22] is depicted in Fig. 15. All the images in the training dataset were utilized for the comparison. Table XII reports the average values of the segmentation performance parameters. The proposed method used to separate touching cells achieves good segmentation accuracy compared to the state-of-the-art techniques when the blood smear image contains multiple cells. The comparison of the classification accuracy achieved by the proposed system with the state of art methods used to classify NRBC is tabulated in Table XIII. The comparison of the proposed segmentation method with the state-of-art methods used to segment the blood smear image and diagnose AML is shown in Table XIV.

TABLE XII
Comparison of Average Values of WBC Segmentation Performance Parameters

Region	SSIM	BF Score	DC	JC	P_C	R_C	Method
Nucleus	**95**	**90**	**96**	**98**	**97**	**99**	**Proposed**
Cell	**95**	**93**	**90**	**97**	**95**	**90**	**Proposed**
Nucleus	73	49	61	53	52	56	[22]
Cell	65	36	58	52	49	53	[22]
Nucleus	92.6	88	95.6	92	92	97	[24]
Nucleus	86	75.2	88.6	81.8	88.5	93	[30]
Nucleus	85.87	65.2	87.9	78.98	87	80.6	[31]
Nucleus	92.6	90.9	92.8	89.7	85.8	93	[32]
Nucleus	91.9	77.9	94	87.47	91.9	97.4	[20]
Cell	84.22	78.08	83.28	81.63	81.9	88.7	[20]

TABLE XIII
Classification and Segmentation Accuracy (in Percent) Comparison of the Proposed Method With the State-of-the-Art Techniques Used to Detect NRBC

Method	Dataset	Images	Segmentation Accuracy	Classification Accuracy	Ref
Convolution Neural Network	Laboratory of Leukemia Diagnostics,Munich University Hospital	18,365	-	Precision= 75 & Sensitivity=87	[42]
FCM clustering algorithm	Midnapur Medical College and Hospital and Medipath Laboratory, Midnapur	250	-	99.42	[43]
Proposed Method	**KMC & PBC dataset**	**1000**	**99.81**	**99.9**	-

TABLE XIV
Comparison of Classification Accuracy Achieved by the Proposed System With State-of-the-Art Techniques in Diagnosing AML

Segmentation method	Dataset	Images	Classifier	Accuracy	Ref
Global thresholding	American society of hematology (ASH) image bank	240	GA-SVM	98.5	[33]
Color clustering	ASH	80	SVM	98	[34]
Color clustering	Shariati Hospital pathology laboratories	330	Multi-SVM	87	[1]
Color clustering	ASH	90	SVM	98	[35]
Color clustering and HMRF method	H.Lee Moffitt Cancer Center	61	-	97.08	[36]
Global threshold	Department of Haematology in the Universiti	301	-	82.9	[37]
Active contour	Sardjito Hospital, Yogyakarta	99	Momentum Backpropagation ANN	93.56	[39]
Fuzzy C-Means	ASH	80	Neural Network	98	[41]
Multi-Otsu thresholding	Cancer Imaging Archive	18,365	Random Forest	93.45	[38]
Proposed Method	**KMC & PBC Dataset**	**1000**	**Random Forest**	**99.48**	-

VI. Conclusion

In this article, a model for detecting acute myeloid leukemia using peripheral blood smear images is presented. A robust method for the detection of nuclei using the K-medoids algorithm is introduced. A novel approach to extract the WBC cytoplasm of cells with various abnormalities without involving manual intervention or cropping is described. The model successfully differentiates the NRBCs from the surrounding WBCs leading to an accurate count of WBCs. The detection of multiple lobes, granules, and Auer rods is accomplished, which helps classify AML and its substages. The model achieved an accuracy of 99.81% in differentiating between NRBCs and WBCs. It also succeeded in differentiating the overlapped erythrocytes from the surrounding WBCs using the novel centroid computation approach based on the position of WBC nuclei. Shape features, color features, and texture features are extracted from the WBCs and NRBCs obtained. Feature selection is performed using the information gain and ranker search method to eliminate features that played less role in classification. The random forest, decision tree, Naive Bayes, and KNN algorithm are chosen for the classification of AML instances, and their classification accuracy is compared. The random forest algorithm is selected for prediction as it has the highest accuracy. The random forest achieved a classification accuracy of 99.48% and a prediction accuracy of 99.2%. The promising results showed that this model could be used as a secondary decision tool by hematologists in their everyday practice. It would help in achieving improved diagnostic accuracy and decrease human error. Additional features can be extracted to capture vacuoles present in the cytoplasm to improve the classification accuracy in future work.

VI. Compliance With Ethical Standards

The authors declare that they have no conflict of interest.

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